

FORWARD DEPLOYED AI ENGINEERING

Capstone Project

Stage One: Discovery

The workbook you complete before you design anything

Fill this in first. Nothing in the later stages is valid unless it can be traced back to something you recorded on these pages.

WEEK ONE · DAYS ONE TO THREE

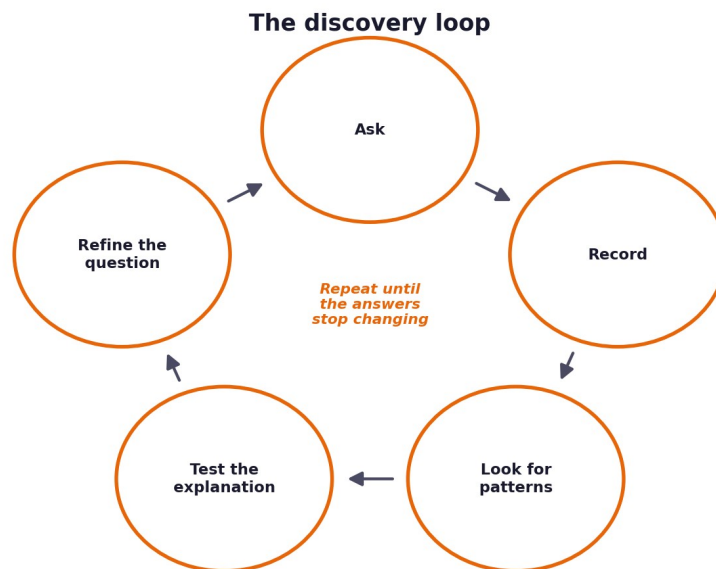
How to use this workbook

This workbook is a set of tables for you to complete. The prompts in the left-hand column tell you what belongs in each row; the remaining columns are yours to fill in. Where a table has blank rows at the bottom, add as many more as you need.

Work through the sections in order. Section one establishes who you are talking to, sections two to five gather the evidence, and section six is where you turn that evidence into a problem statement. Do not attempt section six before the others are genuinely finished, because the whole point of the exercise is that the conclusion follows from the evidence rather than the other way round.

What finished looks like

You are finished with discovery when a further interview tells you nothing you did not already know, and when every claim in your problem statement can be pointed at a specific row in one of these tables. If either of those is not yet true, you are not finished, however much time has passed.



You are finished when a new interview tells you nothing you did not already know.

Figure 1. Discovery repeats until the answers stop changing.

1 Section one: what the interviews tell you

Your discovery evidence is in two places: the five stakeholder interview transcripts in the datasets folder, and the ticket data itself. Start with the transcripts. Read all five before you write anything, then come back and fill in the table below.

The prompts in the left column are the people you spoke to. For each, record what you were trying to learn from them, what they actually told you, and, importantly, what they seemed not to know about their own operation. That last column is often where the useful material is, because each of the five can only see their own part of the problem.

Who	What they told you	What they appear not to know	What you want to verify in the data
Marcus Adeyemi, Head of Support			
Sofia Restrepo, Tier One Agent			
Daniel Okonkwo, Tier Two Engineer			
Ines Varga, Technical Writer			
Ravi Menon, Customer			

Where the accounts disagree

At least three points of disagreement run through the transcripts, and each can be settled by going to the ticket data rather than by deciding whose account sounds more convincing. Identify them, then resolve them with evidence.

The disagreement	Who says what	How the data settles it	What follows from the answer

What nobody said

Real stakeholders leave out the things so familiar to them that they no longer notice them. Record anything you expected somebody to raise that none of the five actually did.

What was never mentioned	Why you would have expected it	How you will check it

2 Section two: what the ticket data shows

Before you interpret anything, count it. Load `development_tickets.json` and fill in the table below from the data itself rather than from impression. Where a figure surprises you, write the surprise down; those are usually the most productive threads to pull.

What to measure	Your figure	Where the figure came from	What surprised you about it
Total tickets in the sample			
Split by channel			
Split by intent category			
Split by urgency			
Proportion marked resolved on first contact			
Average satisfaction rating			
Most frequent single question			
Proportion answerable from existing documentation			
Proportion marked non-fluent, and their outcomes			
Outcomes by customer tier			
Proportion that are repeat contacts			

Two rows matter more than the rest

The proportion of tickets already answerable from existing documentation tells you whether CloudServe lack answers or merely lack a way of reaching them. Those are different problems with different solutions. The non-fluent segment matters for a different reason: something in the transcripts points at it, and the data will tell you whether that person was right.

3 Section three: where the time actually goes

Understanding volume is not the same as understanding effort. A category that accounts for five per cent of tickets but forty per cent of agent hours is where the real problem lives. Work out, as far as your evidence allows, how effort is distributed.

Ticket category	Share of volume	Estimated share of effort	Why the two differ	Evidence for your estimate

The steps an agent takes on a typical ticket

Sofia describes her working process in some detail in the second transcript. Reconstruct it as a sequence of steps and record roughly where the time goes at each one. The steps that sound trivial are frequently where the minutes accumulate.

Step	What the agent does	Roughly how long	Could this be automated? Say why or why not
1			
2			
3			
4			
5			

4 Section four: what the client counts as success

You cannot improve a number nobody is watching. Marcus names the figures he reports and is explicit about which one he actually cares about, and they are not the same. Record what CloudServe measure, who watches each measure, and what would have to change for this project to be considered worthwhile.

Measure	Who watches it	Current value	What they would call success	How confident are you in this?

The sentence test

Complete the following sentence using only figures you recorded above. If you cannot complete it, your discovery is not finished.

Prompt	Your answer
This project will have been worth doing if, within three months of launch, ...	
We will know it did not work if we see ...	
The measure the client will actually be judged on internally is ...	

5 Section five: data, constraints and risk

Everything you build depends on data that already exists, and every piece of that data carries constraints. Record what exists, what condition it is in, and what you are not permitted to do with it. Ines and Daniel both raise concerns about data quality that belong in this table.

Data source	What it contains	Quality and gaps	Access constraints	Private data present?
Support tickets				
Documentation				
Past resolutions				
Customer records				

Initial risk register

List what could go wrong once this system is speaking to customers without supervision. You will develop these further in stage five, but the first pass belongs here, while the problem is still fresh and you have not yet become attached to a particular design.

What could go wrong	How likely	How bad	Who it affects	First thought on preventing it
The system answers confidently and incorrectly				
Private information appears in a reply				
Some customers get consistently worse answers				
The documentation the system relies on goes out of date				

6 Section six: the problem statement

Now, and only now, write down what the problem actually is. Every clause of what you write must be supported by a row you have already filled in, and you are asked to cite those rows explicitly. This is the paragraph your entire project rests on.

Element	Your statement	Which section and row supports it
What the client asked for		
What the evidence suggests they actually need		
The gap between those two		
Who is affected and how		
What will change if this is solved		
What is explicitly not in scope		

The one paragraph version

Write the problem in a single paragraph, in language the head of support would recognise and agree with. No technical vocabulary. If you cannot express it without mentioning retrieval or embeddings, you are describing your solution rather than their problem.

Problem statement

Before you move to stage two

Read your problem statement back against each of the five transcripts in turn and ask whether that person would recognise it as a description of their situation. If any of the five would object, either your framing is off or you have found something they cannot see, and you should be able to say which. A statement that all five would simply agree with is usually one that has not gone far enough.